

Intermediate Project Report

infrared.city-1: Tree detection

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1 Introduction

This section sets the context for the research by outlining the underlying motivation, the proposed methodological approach, and the boundaries of the study. It guides the reader from the broader societal challenge of urban climate adaptation to the specific technical implementation of tree detection using satellite imagery.

1.1 Motivation and Project Overview

As urban areas expand and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims at taking the first step in the direction of better tree coverage in urban areas by creating a model that is able to predict precise tree locations using satellite imagery of densely populated areas.

This research builds upon the foundation laid by previous projects conducted in collaboration with the partner company and environmental simulation platform *infrared.city*. A recent flood risk analysis developed for the platform highlighted that surface impermeability is a dominant factor in urban flooding [3]. However, the authors noted that the lack of detailed, highly localized environmental data limits the precision of urban risk analyses [3]. Concurrently, a second project successfully mapped general urban green spaces using Random Forest models on Sentinel-2 data [4]. While effective for broad classification, that project grouped all vegetation into a single category and reported that initial attempts to use U-Net architectures for detailed spatial analysis failed to yield good results [4]. The authors explicitly recommended future work to revisit U-Net architectures and to distinguish vegetation types more effectively [4].

Recently, there has been a rapidly growing body of research on urban tree detection using deep learning models. One of the most recent papers is by Gong et al. [5], who use satellite data not only to detect trees but also to estimate their size and density. Our project tries to build on both this literature and the identified shortcomings of the previous *infrared.city* projects. By integrating multi-season imagery and temporal attention algorithms, we aim to overcome the previous limitations of the U-Net model and achieve a more granular, object-specific classification than general green spaces.

1.2 Proposed Approach and Data

To achieve reliable urban tree detection, our approach combines multi-spectral satellite imagery with a deep learning architecture optimized for image segmentation.

We utilize publicly available Sentinel-2 satellite images, providing a 10-meter spatial resolution across four color channels: red, green, blue, and near-infrared. These images are retrieved via the API of the Copernicus Programme and are provided by the European Space Agency [6]. A key technical focus of our data strategy is the inclusion of imagery from different seasons to account for the significant impact of seasonality on tree appearance. To provide ground truth for training and testing, we integrate the publicly available “Baumkataster” dataset, which contains exact locations for every registered tree in the city of Vienna [7].

Regarding the model, we employ a U-Net architecture, a specialized Convolutional Neural Network (CNN) consisting of symmetric contracting and expanding paths [8]. While a U-Net implementation in a previous project did not yield promising results, this paper aims to build upon that foundation as the architecture is considered highly accurate for detecting objects in satellite data [9, 10]. Our model extracts features from seasonal multi-channel imagery through its contracting path and utilizes an expanding path for upsampling to predict precise locations.

To improve upon previous attempts, our implementation incorporates skip connections and a temporal attention algorithm to prioritize more meaningful images within the time series [10]. The model performance is optimized using Binary Cross-Entropy loss, comparing our predictions against the split training and test sets derived from the Vienna “Baumkataster”.

1.3 Scope and Limitations

The model in this paper is trained only on satellite images of urban areas and is therefore only supposed to be used for tree detection within city environments. Additionally, the model focuses on tree detection in the temperate climate zone, as this is the vegetation it is trained to observe. Further analyses on the impact of trees on, for example, heat regulation in cities and flooding prevention are out of the scope of this project and could be the topic of future research.

Since this paper uses Sentinel-2 satellite images with 10-meter spatial resolution, issues could arise, as the images might not have a high enough resolution to detect trees reliably. Future work could therefore also focus on improving image quality by using better available satellite data.

2 Related Work

The detection and classification of urban trees using remote sensing imagery is a rapidly evolving field, characterized by a transition from traditional algorithms to advanced machine learning techniques. A comprehensive overview is provided by Zheng et al. [11], who review various methods for individual tree crown detection and delineation from optical imagery, highlighting the growing reliance on deep learning architectures. Building on this methodological shift, Durgut et al. [12] demonstrate the benefits of integrating traditional image processing methods with modern deep learning approaches to further enhance the accuracy of tree crown detection in satellite images.

A significant portion of recent research focuses on highly detailed classification tasks utilizing very high-resolution (VHR) imagery. For instance, Sicard et al. [13] explore object-based classification techniques to accurately distinguish different urban plant species. Similarly, Wenger et al. [14] focus on improving urban tree species classification by combining VHR satellite data with various machine learning models. While these approaches achieve high precision at the individual tree level, they often depend on commercial or less frequently updated satellite data, which can limit the scalability and temporal analysis required for continuous urban climate monitoring.

Addressing the need for scalable and frequently updated data, recent studies have increasingly adopted Sentinel-2 imagery for forestry and urban green space applications. Notably, Freudenberg et al. [15] introduce a dedicated Sentinel-2 machine learning dataset specifically designed for tree species classification in Germany. Their work underscores the viability of using freely available, multi-spectral satellite data for large-scale vegetation mapping.

Furthermore, recent advancements highlight the critical role of multi-temporal analysis in vegetation mapping. Because the spectral signature of trees changes significantly throughout the year due to phenological cycles, relying on single-date imagery often leads to misclassifications. Grabska et al. [16] demonstrate that incorporating the entire Sentinel-2 time series—capturing leaf-on, leaf-off, and flowering periods—greatly enhances the ability of models to differentiate vegetation types. Integrating these temporal signals into deep learning architectures, such as through convolutional temporal attention networks, allows models to prioritize the most meaningful seasonal features for precise identification [10, 16].

Overall, while existing literature predominantly focuses either on VHR imagery for exact crown delineation or on traditional models for general species classification, our project bridges these areas. By leveraging the temporal resolution of Sentinel-2 data [15, 16] combined with a temporal attention U-Net architecture [10], this study aims to provide a scalable and robust solution for urban tree detection that does not rely on costly VHR imagery.

3 Data Sources

This project uses three main kinds of data: tree cadastre records [7] [17] [18] [19], Sentinel-2 satellite imagery [20], and OpenStreetMap (OSM) urban features [21]. Together, these sources support the project’s goal of mapping urban tree canopy from freely available geospatial data. They have already been gathered, cleaned, and stored in reusable formats using two Jupyter notebooks [22] [23], so they can later be used for exploratory data analysis and modelling. The data sources and their cleaning processes are described in the following subchapters.

3.1 Tree cadastre data

Four cities are included: Vienna [7], Paris [17], Barcelona [18], and Hamburg [19]. These datasets were loaded from local raw-data files in different formats. The cadastre sources are important because they provide point-based information about known trees. They are the main direct source of tree locations in this project.

The data preparation follows several steps: first, duplicates are checked, next, columns that are not needed for modelling are removed, while the most useful variables are retained, such as geometry, tree species, height, circumference, crown diameter, and a city identifier. We also standardize coordinate systems and column names to make the datasets comparable across cities. Missing values are handled differently depending on the city. Vienna, Paris, and Barcelona do not contain missing values in the retained fields after cleaning. In the Hamburg Port data, crown diameter and circumference still contain missing values after the first cleaning step. Instead of dropping these rows, the missing values are imputed using the median within the same species, with the overall median used as a fallback. Unknown species names are replaced with the placeholder value "unknown".

After cleaning, the cadastre data are saved in GeoPackage and also Parquet, for compact storage. Barcelona is saved only as Parquet, since its processed version is not stored as a GeoPandas geometry dataset in the same way as the other city datasets.

3.2 Sentinel-2 satellite imagery

For Sentinel-2 data collection, we first considered the Copernicus API [6], but then switched to Google Earth Engine [20] and created monthly median composites from Sentinel-2 images filtered by cloud coverage, which makes the process more reproducible and reduces cloud-related problems compared with manual image selection in the previous project. Google Earth Engine also allowed us to restrict the data retrieval to the exact area used in the tree cadastre data.

The project uses seasonal satellite imagery for spring, summer, and autumn, mainly April, August, and November. These months were chosen to capture different stages of vegetation development. In Paris, October was used instead of November because no suitable November image was available. The image collection is filtered by area of interest, date, and cloud coverage threshold of 20 percent. A stricter 10 percent threshold was tested first, but it did not provide usable images for certain months. From Sentinel-2, we keep four bands: B02, B03, B04, and B08 [24] corresponding to blue, green, red, and

near-infrared. In addition, three vegetation indices are computed: NDVI, EVI, and SAVI [25]. The result is a cleaned multi-season spectral stack for each city.

3.3 OpenStreetMap urban features

We also collect urban context layers from OpenStreetMap. These data come from Geofabrik downloads [21] in GeoPackage format. Separate regional files are used for Austria, Île-de-France, Catalonia, and Hamburg. From each source, we read four thematic layers: roads, buildings, land use, and water.

After loading the OSM layers and converting them to EPSG:4326 [26], they are clipped to the same city bounding boxes derived from the cleaned tree-cadastre data. This keeps the cadastre data, Sentinel-2 imagery, and OSM layers spatially aligned within the same study areas. At this stage, the OSM data are only lightly cleaned. We also check geometry validity and empty geometries, but no strong filtering of feature classes is applied yet to preserve flexibility for later modelling. The cleaned layers are saved for reuse in GeoPackage and Parquet format and are prepared for exploratory data analysis.

4 Methodology

For this tree detection task we implemented a U-Net regression architecture with a temporal attention mask which predicts the number of trees per pixel of Sentinel-2 data where the prediction labels are derived from communal tree registries.

4.1 Model

Former papers suggest that for image segmentation using satellite data the U-Net deep learning model is the most accurate for detecting objects, for example street detection [9] as well as crop field detection [10]. Furthermore, last year’s *infrared.city* project also tried implementing a U-Net architecture but failed to yield promising results, also the data coaches deem it a well-fitting model.

The U-Net model is a special type of Convolutional Neural Network and consists of a contracting path and an expanding path. First, the satellite image is put through an input convolution so it has the correct structure for further processing. It is then passed into the contracting path to down-sample the data in the convolution and pooling layers, which are used to extract features from the original imagery. Our model uses four convolutional layers where during contraction for each layer the image size gets smaller due to pooling meanwhile evermore features are extracted from the images. The same process happens vice versa for the expansionary path. The expanding path consists of the same structure of convolution and pooling layers but this time used to up-sample the before contracted imagery [8]. To get the final prediction of the number of trees per pixel another convolutional layer is deployed. As suggested in similar work this project will use skip connections between the expanding and contracting path as well as a temporal attention algorithm to prioritize more meaningful images [10] and take advantage of the seasonality and changing vegetation of trees. The model uses smoothed L1 loss (also called Huber loss) and the tree registry data as a ground truth.

Figure 1 presents the BEAM representation of the proposed tree detection model

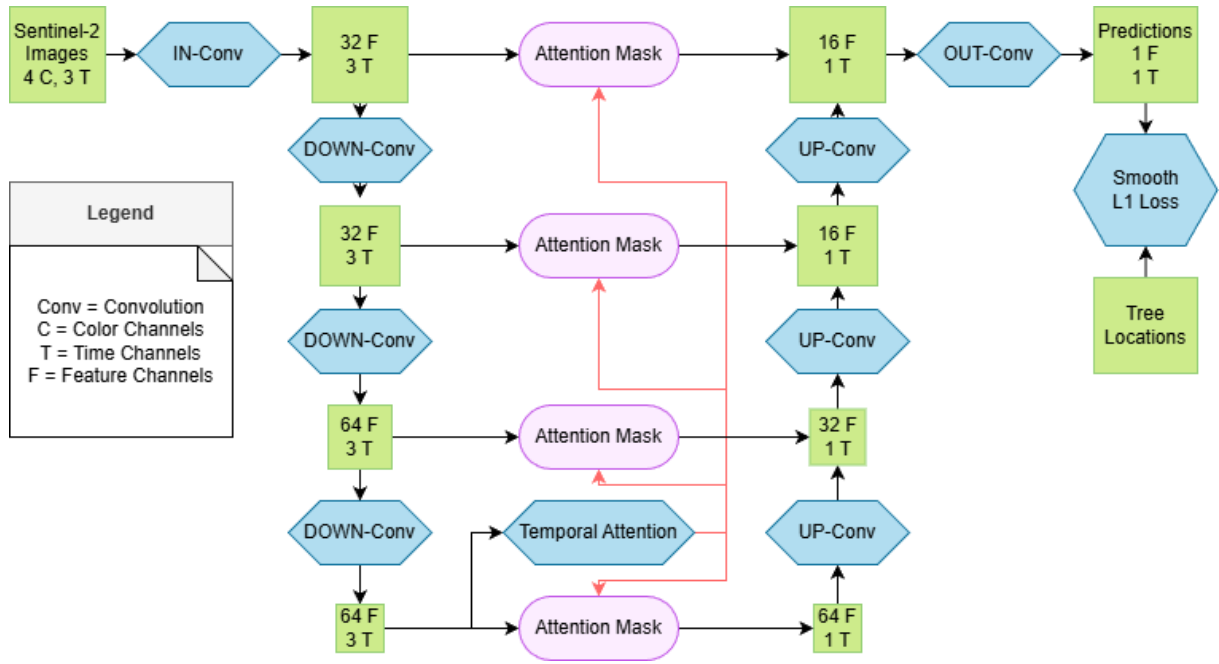


Figure 1: BEAM model of the proposed tree detection model

5 Implementation

This chapter describes the workflow and hyperparameters used for data preprocessing and model training.

5.1 Tools

This model was implemented using the programming language python and the deep learning library pytorch. To store the data in an easily machine-readable format suited for machine learning tasks the library numpy was used and the data stored as numpy arrays. For data import and preprocessing the libraries rasterio and pyproj were used.

5.2 Data Preprocessing for Model

In order to get the data into a usable shape for the U-Net model the following preprocessing steps were applied.

The Sentinel-2 images are patched into smaller 256 pixels wide quadratic images and a stride of around 128 pixels is used so that the final patches are overlapping and more training data can be extracted from the images. The four color channels red, green, blue and near-infrared are used and the patches from different months of the year but from the same location are stacked on top of each other. We used data from April, August and November (except for Paris where data from November was too cloudy and we had to use October). The satellite imagery was transformed into structured numerical arrays, resulting in a data representation with the dimensions $N \times T \times C \times H \times W$, where N denotes the number of image patches, T the number of temporal observations (months), C the number of spectral channels, and H and W the spatial dimensions (height and width) respectively.

For the prediction labels patches at the same geographical locations as the Sentinel-2 patches were created and filled with the number of trees per pixel. This data was derived from the local tree registries which provide the exact coordinates of the trees in the corresponding city. The data was then split into a train, validation and test sample.

5.3 Model Training, Testing and Hyperparameters

The model was built according to the description in the chapter methodology, although the depth of the model is limited due to computational restraints. At the bottleneck where the feature space is the richest and the attention algorithm is deployed the U-Net had 64 feature channels, before down-sampling 32 and after up-sampling 16. The attention mechanism uses 8 attention heads, the QKV vectors have 8 dimensions the embedding vectors 64.

The used learning rate is 0.001 and training ran for ten epochs on around 100 batches each containing four data points. 5-fold cross-validation is used to train five different models. To evaluate the best of these which will be saved for further use and testing the loss, RMSE and R-squared values are calculated. The chosen model was then tested on unseen data from a city which was not used in the training process.

6 Preliminary Results

The preliminary results of this project consist of an exploratory data analysis conducted across all three data sources: the Baumkataster, the Sentinel-2 satellite imagery and the OpenStreetMap (OSM) urban context layers. The goal of the analysis was to assess data quality, learn feature distributions and identify which indicators are likely to be useful for training the U-Net model.

6.1 Baumkataster Analysis

This section contains some analysis on the Baumkataster data sets.

6.1.1 Dataset Overview and Tree Counts

The cleaned Baumkataster data contain 226,998 trees for Vienna, 217,454 for Paris, 233,505 for Hamburg and 43,367 for Barcelona, as shown in Figure 2. Vienna, Paris, and Hamburg are roughly balanced in size and are suitable as training cities. However, Barcelona lacks size attributes and a geometry column and therefore it will only be used as point locations for cross-city generalization testing.

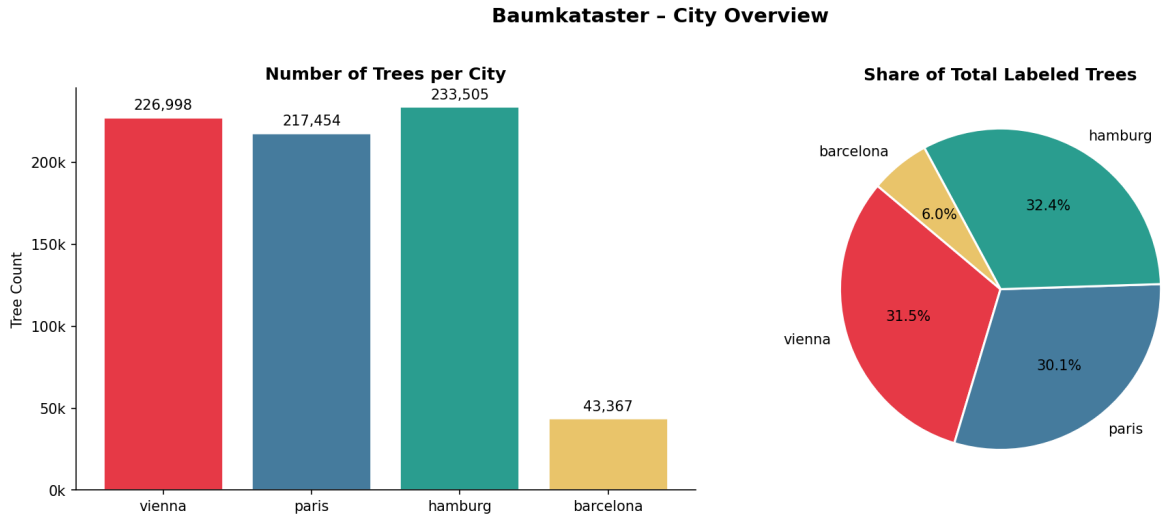


Figure 2: Number of labeled trees per city (left) and their relative share of the total dataset (right).

The available size attributes differ across cities. Vienna provides height categories, trunk circumference, and crown diameter. Paris provides height and circumference. Hamburg provides circumference and crown diameter.

6.1.2 Tree Size Distributions

Figure 3 shows the distributions of height, circumference, and crown diameter across cities. The height histogram reveals a data quality issue specific to Vienna. Its height column stores ordinal category codes from 1 to 8, representing height ranges from 0–5 m up to >35 m, rather than continuous metre values. This is confirmed by the raw mean of 2.21, which is not interpretable as metres but corresponds to the most frequent category (6–10 m, midpoint 8.0 m). The planned fix is to decode these codes to midpoint values

before model training. Paris height values, by contrast, are stored in metres and show a realistic distribution with a median around 8 m. The circumference distributions (same problem for Vienna)...

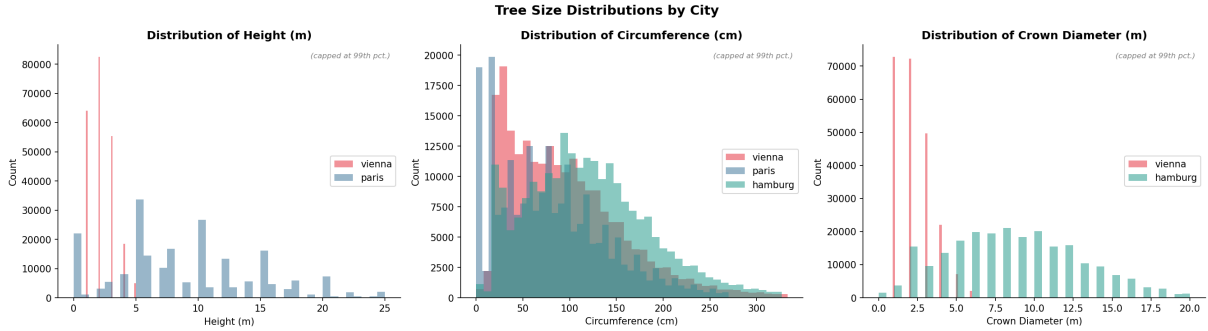


Figure 3: Distributions of height, trunk circumference, and crown diameter by city (capped at 99th percentile for readability). Note that Vienna height and crown diameter values are category codes, not metres.

6.1.3 Species Composition

Species diversity is high across all cities, with 670 unique species in Vienna, 996 in Paris, 354 in Hamburg, and 330 in Barcelona. Across all four cities, 86 species are shared, predominantly from the *Abies* and *Acer* genera. Species identity will not be used as a direct model input, but the diversity and climate-specific composition inform expectations about canopy shape and size variation across cities. Figure 4 shows the ten most common species in each city. Vienna is dominated by *Acer platanoides* (Norway maple, 19,244 trees) and *Aesculus hippocastanum* (horse chestnut, 11,661 trees), both classic central European street tree species. Paris is dominated by *Platanus x hispanica* (London plane, 38,113 trees). Hamburg is dominated by *Quercus robur* (English oak, 42,965 trees) and *Tilia x europaea* (European lime, 33,852 trees). Barcelona shows a different composition with subtropical species such as *Tipuana tipu* and *Pinus pinea*, reflecting its Mediterranean climate.

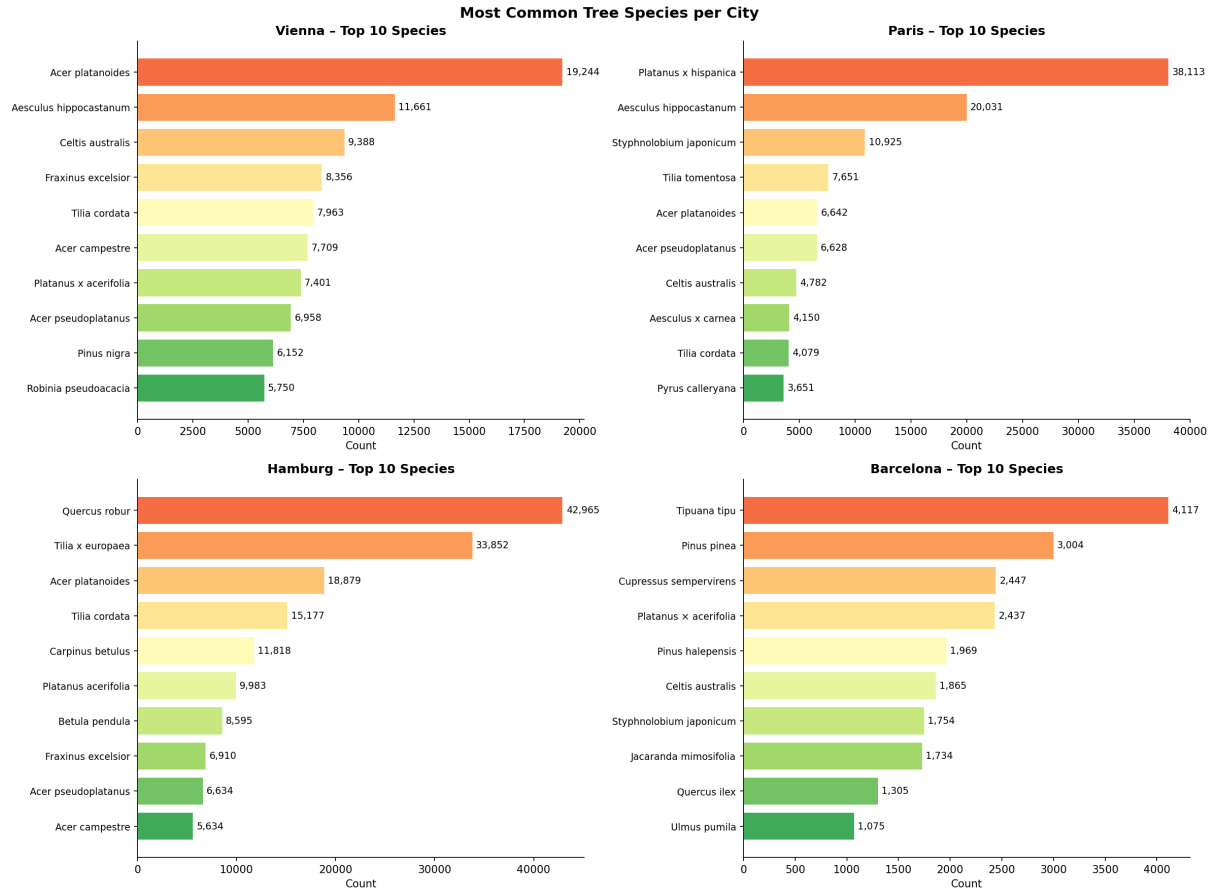


Figure 4: Top 10 most common tree species per city. Species composition differs substantially between climate zones, with Barcelona showing subtropical species absent from the other cities.

6.2 Sentinel-2 Analysis

All 12 city-season combinations were successfully loaded. Band statistics were computed across the seven available channels: B02 (blue), B03 (green), B04 (red), B08 (NIR), NDVI, EVI, and SAVI.

NDVI values are highest in summer (August) across all cities, with Hamburg reaching a mean of 0.42 compared to approximately 0.29 in spring and autumn. This confirms that summer imagery carries the strongest vegetation signal. The seasonal contrast between summer and autumn NDVI is particularly valuable for distinguishing deciduous trees from other permanently green surfaces such as grass or artificial turf.

Another data quality issue was identified with the SAVI indices. EVI values exceed 1.0 for all cities in summer, which is false as EVI is bounded between -1 and $+1$. This is a systematic issue across all cities. The root cause of this is that the index the formulas were applied to the raw pixel values stored in the satellite files, which are large integers, instead of the actual light reflectance values they represent. Converting raw values to reflectance requires dividing by 10,000. NDVI is unaffected here because its formula divides two numbers, so the scale cancels out either way. EVI is more complex and returns incorrect results when the input values are not in the expected range.

6.3 OSM Analysis

The OSM layers were loaded for all four cities across four feature types: roads, buildings, landuse, and water. Road networks are dominated by **footway**, **service**, and **residential** classes across all cities. Hamburg’s OSM extract is noticeably sparser than the other cities, with 24,184 road features compared to 161,378 for Vienna, which will be investigated before using Hamburg as a training city.

Building footprints are present for all cities and will serve as reliable negative labels during training mask generation, as no trees are expected on rooftop surfaces. Road classes such as **footway**, **path**, **steps**, and **track** will not be used as hard negative masks, as these routes frequently pass through parks alongside trees and would introduce false negatives into the training labels.

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